

Non-Iris Proofmode for Temporal Logic (WIP)

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Slides:



Work in Progress!

Feedback appreciated :)

Temporal Logic Primer

Temporal logic focus on proving when / if something happens

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- ▶ E.g.: If a request is made, a response will eventually happen

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- ▶ Next P ($\bigcirc P$): P holds after 1 step of the trace
- ▶ Globally P ($\Box P$): P holds for all suffixes of the trace
- ▶ Eventually P ($\Diamond P$): P holds for some suffix of the trace

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- ▶ $\Box(P \rightarrow \bigcirc \Diamond Q) \vdash \Diamond P \rightarrow \bigcirc \Diamond Q$

Story Time

The need for mechanisation infrastructure for temporal properties

Three Options

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- ▶ Remove “All results have been mechanised in Rocq”
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- ▶ Alas, manually threading trace suffixes is a pain

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Option 3: Using existing libraries

- ▶ Browse selection of existing (LTL) libraries in Rocq
- ▶ Realise they primarily focus on metatheory => not many helper lemmas
- ▶ Additionally, requires manually managing proof context

Goal: Mechanisation Infrastructure for Temporal Properties

MoSeL: Modal Separation Logic (Proofmode)

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Key features

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- ▶ Managed proof context for spatial and persistent propositions
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Infrastructure based on modal laws

$\triangleright$ -MONO

$$\frac{P \vdash Q}{\triangleright P \vdash \triangleright Q}$$

$\triangleright$ -SEP

$$\triangleright P * \triangleright Q \dashv\vdash \triangleright(P * Q)$$

$\triangleright$ -INTRO

$$P \vdash \triangleright P$$

MoSeL Demo

Key Idea: Instantiate (and Extend) MoSeL for Temporal Logic

Key Challenges

Retrofitting MoSeL infrastructure

- ▶ Enforces unused step-indexing (later) and resources (sep. conjunction)
- ▶ Requires modal law $\text{True} \vdash M \text{ True}$ for any modality M
- ▶ Does not support all conventional modal laws, e.g. those of eventually $\Diamond$

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Defining “persistently”: $\Box \cdot P$

- ▶ Unclear if there is a canonical choice (alike Iris)
- ▶ Intuition is that P holds “independently” (in Iris, without owning anything)
- ▶ Current choice is temporal logic “globally”: $\Box \cdot P \triangleq \Box P$
- ▶ Alternative choice is $\Box \cdot P \triangleq \lambda_ . \forall tr . P \text{ tr}$

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Defining model of traces

- ▶ Infinite vs possibly-infinite maximal traces

Steps Towards a Temporal Logic Proofmode

Trace Logic Foundation

Possibly-infinite maximal traces over a relation $R : S \rightarrow L \rightarrow S \rightarrow \text{Prop}$

- ▶ $tr \in \text{Trace} \triangleq \langle s \rangle \mid s \xrightarrow{\ell} tr \mid \perp$
- ▶ Bottom element is an absorbing state for terminated traces

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Propositions indexed by traces with expected definitions of usual connectives

- ▶ $P, Q \in \text{tProp} \triangleq \text{Trace} \rightarrow \text{Prop}$
- ▶ $P \vdash Q \triangleq \forall tr. P\ tr \implies Q\ tr$
- ▶ $P \wedge Q \triangleq \lambda tr. P\ tr \wedge Q\ tr \quad P \rightarrow Q \triangleq \lambda tr. P\ tr \rightarrow Q\ tr$

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Persistence is defined as globally

- ▶ $\Box P \triangleq \Box P$

Modalities are inductively / coinductively defined with expected semantics

- ▶ Next: $\bigcirc P \equiv \lambda tr. P \text{ (drop 1 } tr)$
- ▶ Eventually: $\Diamond P \equiv \lambda tr. \exists n. P \text{ (drop } n \text{ } tr)$
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Possibility of finite traces complicates picture

- ▶ Recall required modal laws: $\text{True} \vdash \bigcirc \text{True}$ and $\text{True} \vdash \Box \text{True}$
- ▶ Propositions on successive elements must be defined for singleton trace $\langle s \rangle$
 - ▶ $(\bigcirc P) \langle s \rangle \equiv P \perp \quad (\bigcirc P) \perp \equiv P \perp \quad (\Box P) \langle s \rangle \equiv P \langle s \rangle \wedge (\Box P) \perp \quad (\Box P) \perp \equiv P \perp$
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Modal laws of temporal modalities are partially supported by MoSeL

Modal Laws (an excerpt)

Globally ($\Box$) vs. Iris Persistent ($\Box^\circ$)

$\Box$ -MONO

$$\frac{P \vdash Q}{\Box P \vdash \Box Q}$$

$\Box$ -AND

$$\Box P \wedge \Box Q \dashv\vdash \Box(P \wedge Q)$$

$\Box$ -TAUT

$$\text{True} \vdash \Box \text{True}$$

$\Box^\circ$ -EXISTS

$$\Box^\circ \exists x. P \not\vdash \exists x. \Box^\circ P$$

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Eventually ($\Diamond$) vs. Iris Update ($\triangleright$)

$\Diamond$ -MONO

$$\frac{P \vdash Q}{\Diamond P \vdash \Diamond Q}$$

$\Diamond$ -IDEMP

$$\Diamond P \dashv\vdash \Diamond \Diamond P$$

$\Diamond$ -AND-L

$$\Diamond(P \wedge Q) \vdash \Diamond P \wedge \Diamond Q$$

$\Diamond$ -AND-R

$$\Diamond P \wedge \Diamond Q \not\vdash \Diamond(P \wedge Q)$$

Temporal Logic Proofmode Demo

Reflections on Definition of Persistent

Current definition of persistent ($\Box P \triangleq \Box P$) is incompatible with MoSeL

- ▶ MoSeL requires invalid law: $\Box \exists x. P \not\vdash \exists x. \Box P$
- ▶ But law might not be strictly required; mostly used in step-indexing

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Alternative definition ($\Box P \triangleq \lambda_. \forall tr. P \ tr$) is unsatisfactory

- ▶ Does not let us leverage MoSeL infrastructure for persistent
- ▶ Requires treating $\Box$ as a separate modality
- ▶ Incompatible with current tooling for $\Diamond$
- ▶ Yields weaker logic-internal fixpoint theorems

More than just Linear Temporal Logic (LTL): Fixpoints

We have least (μ) and greatest (ν) fixpoints for any function monotone in $\Box$

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- ▶ $\Diamond P \triangleq \mu X. P \vee \bigcirc X$
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Fixpoint theorems inherit choice of $\Box$

- ▶ Theorem: $\Box((P \vee \bigcirc Q) \rightarrow Q) \vdash \Diamond P \rightarrow Q$
- ▶ Stronger $\Box$: more fixpoints, weaker theorems (e.g. $\Box P \triangleq \forall tr. P \text{ tr}$)
- ▶ Weaker $\Box$: less fixpoints, stronger theorems (e.g. $\Box P \triangleq \Box P$)

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But unclear if there is a good use case for this

- ▶ Iris-style abstract predicate use case seem to be about abstracting permissions
- ▶ We have yet to come up with a good example
- ▶ Ideas welcome :~)!

To Conclude

We Need Help!

Settling on a definition of persistent

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- ▶ Alternate does not allow us to leverage MoSeL persistent context for globally

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- ▶ Separation logic resources?
- ▶ Further generalisation of MoSeL modality infrastructure?

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So far mostly an engineering project..

- ▶ Experiences in pitching proofmode projects appreciated

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Modal temporal logic has correspondences to Iris-style reasoning

- ▶ Foundations in similar modal laws
- ▶ Albeit with key distinctions

$$\vdash \Box(\textit{Question} \rightarrow \Diamond \textit{Answer})$$

Rocq code available at:



Expected Contributions

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Insights into further generalisation of MoSeL (or modal proofmodes in general)

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First(?) steps towards temporal separation logic

- ▶ Conceptually stimulating
- ▶ Practically unclear what it even means